3-4

Date time: March 4th at 12:00
Presenter: Dr. Perry Alexander
Presentation Name: Research

Summary

Dr. Alexander begins his speech by talking about his daily routine and his personal workspace. He then gives an excellent quote that describes the nature of what he does by John von Neumann. While he only spent a few minutes talking about the quote (something along the lines of "you do not understand things, you just get used to them”), it was really impactful and something that I have been thinking about all day. I have a deep obsession with understanding everything so this quote made me question whether I want to pursue a PhD or not (though it made me want to pursue one even more).

Dr. Alexander then moves into the core philosophy of his work. He described research as a process of "active learning" which is different from traditional classroom education. He uses the metaphor of "jumping into a hole" to explain the research experience, where students find something they are interested in and must navigate a space with no clear exit or established rules. He argues that research is essentially education in its purest form, where the student eventually surpasses the advisor to become a global expert in a specialized niche.

Following this, he outlines the various trajectories for graduate students. One thing he mentioned was how most students don't stay in academia. He mentions that there are many opportunities in government laboratories like the NSA, large industrial labs such as Google and Microsoft, and the high-risk, high-reward world of startups. He shares his own experience with starting companies to demonstrate how academic theory can be translated into commercial technology. He also explains the differences between master’s degrees and PhDs and explains why he thinks a thesis master's is better than a coursework master's.

In the final portion of his talk, he explains what being a researcher and a professor is like. It includes heavy amounts of writing, presenting, and collaborative "board work" rather than just solitary coding. He ends the speech by talking about Artificial Intelligence and the anxiety it is causing. He says AI as an interesting but highly inefficient "token-guessing" tool and encourages his audience not to fear it, but to view it as another technological shift (he compared it to the .com era) that requires human creativity and resilience to master.

What I learnt

Because of the presentation, I have a much better understanding of what research really is like. One of the fears that I had was that following a PhD, my job will be limited to academia, a field I really don't want to stay in. Another fear was that a PhD is isolating and involved a lot of coding

on your own with not much help. Both of these things were far from the truth. In his presentation, Dr. Alexander mentioned that most of his students ended up getting a job outside of academia like at Galois or big tech research. Additionally, Dr. Alexander's day seemed extremely collaborative and made me wonder when he actually has time to write code.
Another important thing I learnt was the concept of active and passive learning explained by Dr. Alexander. The main difference is that active learning is when you have to go out of your way to learn a specific thing while passive learning is you are taught a thing that might not be specific to what you want to learn. I looked deeper into this concept, and it turns out you are much more likely to remember things that you use active learning for and that a good study habit is to combine them by taking what you learn in class a step further.

What I will do different.

As I mentioned right before, I will change my study habits to be a bit more active. Instead of just learning what we are supposed to for an exam, I will try to take it one level deeper, either an application of the topic or something that builds off it. This will use more active learning and help me understand and remember the topic better. More importantly though, this will give me a taste of the topic to see if I like it. If I do, I can enter an aforementioned "rabbit hole". This will give me more view of what a PhD is like which will help me decide my future. Additionally, the von Neumann quote had me thinking whether I would want to pursue a PhD and the answer is a resounding yes. I want to get comfortable with complex topics and be one of the best in a field.